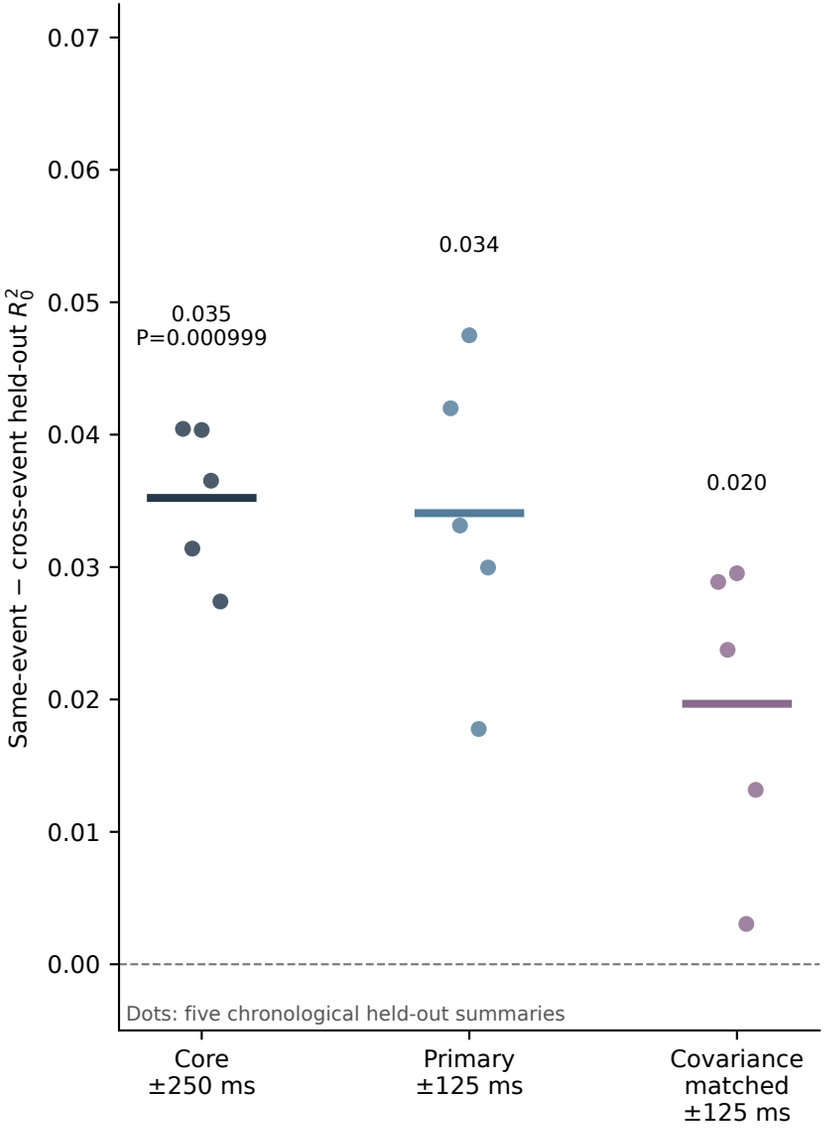
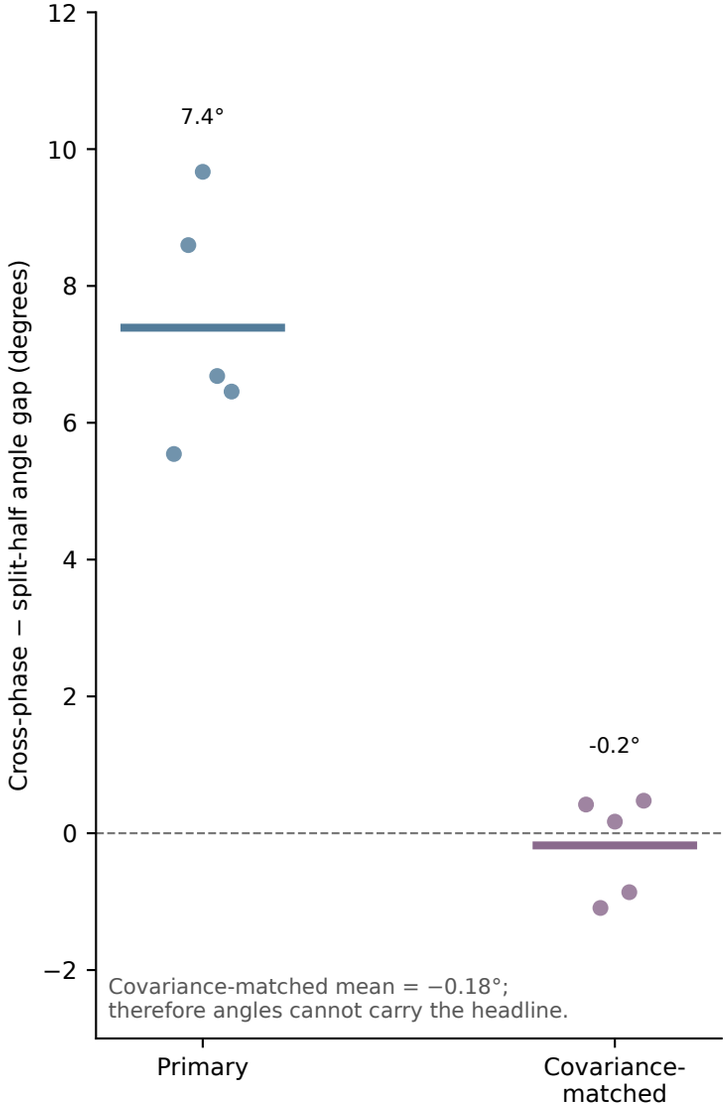
Event-specific held-out predictability is robust; geometric rotation is not established

a Predictive specificity survives covariance matching



b Axis rotation is not resolved



c Evidence boundary

SUPPORTED

- ✓ Same-event maps predict held-out activity better than cross-event transfer.
- ✓ The advantage persists after third-area, paw, task-time and covariance controls.
- ✓ All five chronological held-out summaries are positive.

NOT RESOLVED

- × The fitted axes themselves are not reliably shown to rotate.
- × No perturbation: not causal routing, information flow or anatomical direction.
- × One mouse/day: no animal-level generalization.

Frozen outputs only. Covariance matching uses training-only Ledoit–Wolf whitening and pooled-covariance recoloring. Exact four-event task-time matching was infeasible because the four events had no common time support.